

**13.5 / 24**

MARKS GRADED

All answers read and graded.

4 knowledge gaps

1 careless slip

● correct ● partial ● lost ● pending**1**

1 mark

1 / 1

The HCF of 6 and 20 is:(a) 1 (b) 2 (c) 4 (d) 120

STEP 1 **Correct**

Identify the HCF of 6 and 20.

(b) 2

↳ Correctly identified the HCF.

2

2 marks

0 / 2Find the roots of the quadratic equation $2x^2 - 7x + 3 = 0$ by factorisation.**STEP 1** **Incorrect** **Concept gap****+0**

State the quadratic equation.

 $2x^2 - 7x + 5 = 0$ ↳ The quadratic equation was miscopied from the question. The original question was $2x^2 - 7x + 3 = 0$. This is a departure from the question.✓ $2x^2 - 7x + 3 = 0$ **STEP 2** **Incorrect****+0**

Splitting the middle term.

 $2x^2 - 2x - 5x + 5 = 0$

↳ This step is correct for the miscopied equation, but no marks are awarded as it is after the departure.

✓ $2x^2 - 6x - x + 3 = 0$ **STEP 3** **Incorrect****+0**

Factorising by grouping.

 $2x(x - 1) - 5(x - 1) = 0$

↳ This step is correct for the miscopied equation, but no marks are awarded as it is after the departure.

✓ $2x(x - 3) - 1(x - 3) = 0$ **STEP 4** **Incorrect****+0**

Writing the factors.

$$(2x - 5)(x - 1) = 0$$

↳ This step is correct for the miscopied equation, but no marks are awarded as it is after the departure.

$$\checkmark (2x - 1)(x - 3) = 0$$

STEP 5 **Incorrect**

+0

Finding the roots.

$$x = 5/2, x = 1$$

↳ The roots are correct for the miscopied equation, but no marks are awarded as it is after the departure.

$$\checkmark x = 1/2, x = 3$$

EXAMINER NOTE

From this step on you were solving a different equation from the one set — check each line against the question as you go.

3 3 marks

3 / 3

Find the sum of the first 20 terms of the AP: 3, 7, 11, 15, ...

STEP 1 **Correct**

+1

Identify the first term (a), common difference (d), and number of terms (n).

$$a = 3, d = 7 - 3 = 4, n = 20$$

↳ Correctly identified a, d, and n.

STEP 2 **Correct**

+1

Write the formula for the sum of n terms of an AP (S_n).

$$S_n = n/2 [2a + (n - 1)d]$$

↳ Correct formula for S_n .

STEP 3 **Correct**

+1

Substitute values and calculate the sum.

$$S_{20} = 20/2 [2 \times 3 + 19 \times 4] = 10 [6 + 72] = 10 \times 78 = 780$$

↳ Correct substitution and calculation.

4 3 marks

0 / 3

If $\sin A = 3/5$, find the values of $\cos A$, $\tan A$ and $\sec A$.

STEP 1 **Correct**

+0

State the given value of $\sin A$.

$$\sin A = 3/5$$

↳ Correctly stated the given value.

STEP 2 **Incorrect** **Concept gap**

+0

Find the value of $\cos A$.

$$\cos A = 1 - \sin A = 1 - 3/5 = 2/5$$

↳ Incorrect formula used for $\cos A$. It should be $\cos^2 A = 1 - \sin^2 A$. This is a conceptual error and a departure from the correct method.

$$\checkmark \cos A = \sqrt{1 - \sin^2 A} = \sqrt{1 - (3/5)^2} = \sqrt{1 - 9/25} = \sqrt{16/25} = 4/5$$

STEP 3 **Incorrect**

+0

Find the value of $\tan A$.

$$\tan A = \sin A / \cos A = (3/5) / (2/5) = 3/2$$

↳ Correct method for $\tan A$, but uses the incorrect value of $\cos A$ from the previous step. No marks awarded as it is after the departure.

$$\checkmark \tan A = (3/5) / (4/5) = 3/4$$

STEP 4 **Incorrect**

+0

Find the value of $\sec A$.

$$\sec A = 1/\cos A = 5/2$$

↳ Correct method for $\sec A$, but uses the incorrect value of $\cos A$ from the previous step. No marks awarded as it is after the departure.

$$\checkmark \sec A = 1 / (4/5) = 5/4$$

EXAMINER NOTE

From this step on you were solving a different equation from the one set — check each line against the question as you go.

5

3 marks

2.5 / 3

Three resistors of $2\ \Omega$, $3\ \Omega$ and $6\ \Omega$ are connected in parallel to a 6 V battery. Find (i) the equivalent resistance of the combination and (ii) the current drawn from the battery.

STEP 1 **Correct**

+0.5

Write the formula for equivalent resistance in parallel and substitute values.

$$1/R = 1/2 + 1/3 + 1/6$$

↳ Correct formula and substitution for parallel resistance.

STEP 2 **Correct**

+1

Calculate the equivalent resistance (R).

$$= (3 + 2 + 1)/6 = 6/6 = 1R = 1$$

↳ Correct calculation of equivalent resistance.

STEP 3 **Partial** **Presentation**

+1

Calculate the current drawn from the battery using Ohm's Law.

$$I = 6/1 = 6$$

↳ Correct calculation of current, but missing the unit (Amperes).

$$\checkmark I = 6\text{ A}$$

6 2 marks

0 / 2

Find the coordinates of the point which divides the line segment joining (2, 3) and (5, 6) internally in the ratio 2 : 1.

STEP 1 Incorrect

+0 -2

Find the coordinates of the point using the section formula.

Don't know

↳ The question was not attempted.

7 2 marks

2 / 2

Balance the following chemical equation and state the type of reaction: $\text{Fe} + \text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$

STEP 1 Correct

+1

Balance the chemical equation.

$\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$

↳ The student initially wrote an unbalanced equation but then correctly balanced it. Full marks for the correct final balanced equation.

STEP 2 Correct

+1

State the type of reaction.

It is a redox reaction.

↳ Correctly identified the type of reaction.

8 2 marks

2 / 2

Why is the wall of the left ventricle of the human heart thicker than that of the right ventricle?

STEP 1 Correct

+1

Explain the function of the left ventricle and its wall thickness.

The left ventricle pumps oxygenated blood to all parts of the body, so it has to pump with more force.

↳ Correctly explained the role of the left ventricle.

STEP 2 Correct

+1

Explain the function of the right ventricle and its wall thickness.

The right ventricle pumps blood only to the lungs. So the left wall is thicker.

↳ Correctly explained the role of the right ventricle and linked it to the left ventricle's thickness.

9 3 marks

2.5 / 3

(i) Two dice are thrown together. Find the probability of getting a sum of 7. [2] (ii) A bag contains 5 red and 3 blue balls. One ball is drawn at random. Find the probability that it is blue. [1]

STEP 1 **Correct**

+0.5

Part (i): State the total number of outcomes when two dice are thrown.

Total outcomes = 36

↳ Correctly identified total outcomes.

STEP 2 **Partial** **Calculation**

+0.5

Part (i): List the favorable outcomes for getting a sum of 7.

Sum 7: (1,6) (2,5) (3,4) (4,3) (5,2)

↳ Missed one favorable outcome (6,1). There are 6 outcomes for a sum of 7, not 5.

✓ Sum 7: (1,6) (2,5) (3,4) (4,3) (5,2) (6,1)

STEP 3 **Correct**

+0.5

Part (i): Calculate the probability of getting a sum of 7.

P(sum 7) = 5/36

↳ Correctly calculated probability based on their (incorrect) count of favorable outcomes. (ECF applied)

✓ $P(\text{sum } 7) = 6/36 = 1/6$ STEP 4 **Correct**

+0.5

Part (ii): State the total number of balls in the bag.

Total balls = 5 + 3 = 8

↳ Correctly identified total number of balls.

STEP 5 **Correct**

+0.5

Part (ii): Calculate the probability of drawing a blue ball.

P(blue) = 3/8

↳ Correctly calculated the probability of drawing a blue ball.

10 3 marks**0.5 / 3**

An object is placed 20 cm in front of a concave mirror of focal length 15 cm. Find the position of the image.

STEP 1 **Partial** **Concept gap**

+0.5

State the given values with correct sign conventions (u, f).

u = 20 cm, f = -15 cm

↳ The object distance 'u' should be -20 cm according to sign conventions for an object placed in front of a mirror. This is a conceptual error and a departure from the correct problem setup.

✓ $u = -20 \text{ cm}, f = -15 \text{ cm}$ STEP 2 **Incorrect**

+0

Write the mirror formula.

 $1/v + 1/u = 1/f$

↳ Correct formula, but no marks awarded as it is after the departure.

STEP 3 Incorrect

+0

Substitute values into the formula.

$$1/v = 1/f - 1/u = -1/15 - 1/20$$

↳ Correct substitution for the student's values, but no marks awarded as it is after the departure.

$$\checkmark 1/v = 1/(-15) - 1/(-20) = -1/15 + 1/20$$

STEP 4 Incorrect

+0

Perform calculations to find $1/v$.

$$= (-4 - 3)/60 = -7/60$$

↳ Correct calculation for the student's values, but no marks awarded as it is after the departure.

$$\checkmark = (-4 + 3)/60 = -1/60$$

STEP 5 Incorrect

+0

Calculate the image position (v) and state its nature.

$$v = -60/7 = -8.57 \text{ cm Image is 8.57 cm in front of the mirror.}$$

↳ Correct calculation and interpretation for the student's values, but no marks awarded as it is after the departure.

$$\checkmark v = -60 \text{ cm. Image is 60 cm in front of the mirror.}$$

EXAMINER NOTE

From this step on you were solving a different equation from the one set — check each line against the question as you go.

Where your marks went

You scored 13.5 of 24 on the work we could read. 4 knowledge gaps (revise the method) and 1 careless slip (slow down and show every step) cost you marks.